The Open Source Audit

Course: Open Source Software
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Software Chosen: Git
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Case Study

When Control Breaks Trust: The Story Behind Git's Creation

A Case Study in Open Source Dependency, Control, and Developer Autonomy

1. Introduction

Open-source software plays an important role in modern computing. Many tools that developers use today are built and maintained by communities rather than single companies. These tools are freely available and can be modified, which allows continuous improvement.

In this project, I have selected Git as the open-source software to study. The aim is not just to understand what Git does, but also why it was created, how it works in a Linux environment, and what values it represents in the open-source ecosystem.

Part A – Origin and Philosophy

A1. The Problem Git Was Created to Solve

Git was not created because developers wanted a better tool.
It was created because they suddenly had no tool they could trust.

In 2005, the Linux kernel project depended on a system called

BitKeeper. It worked well technically, but it came with a hidden cost: control was not in the hands of the developers. When that relationship broke, access was revoked. Overnight, one of the largest open-source projects in the world lost its coordination system.

This is the real origin of Git—not convenience, but dependency failure.

Linus Torvalds did not try to replace BitKeeper with something similar. He rejected the model entirely. Instead of central control, he designed Git around a simple but powerful idea:

> Every developer should have a complete copy of the project.

This changed everything.

Earlier systems assumed a central authority. Git removed that assumption. Each contributor could work independently, verify history locally, and collaborate without relying on a single server. This was not just a technical improvement — it was a shift in power.

Git also prioritized:

- Speed
- Integrity
- Distributed workflow — no single point of failure

But the deeper reason Git succeeded is this: It aligned the tool with the philosophy of open source.

If the code is open, the process managing that code cannot be controlled by a closed system. Git solved not just a development problem, but a structural contradiction.

A2. The License — What It Actually Says

Git is licensed under the GNU General Public License version 2 (GPL v2). This license ensures that the software remains free and open.

The Four Freedoms

- Freedom to run the program
- Freedom to study the source code
- Freedom to modify it
- Freedom to distribute copies

These freedoms ensure that users are not restricted.

Can a Company Modify and Sell Git?

Yes, a company can modify and sell Git. However, they must also share the source code of their modifications. This is because GPL is a copyleft license.

GPL vs MIT

- GPL requires modifications to remain open
- MIT allows modifications to be kept private

If I were to choose a license, I would select GPL if I want to keep software open, and MIT if I want wider usage.

Free as in “Free Beer” vs “Free Speech”

- Free beer → no cost
- Free speech → freedom to modify

Git is free in both ways, but the freedom aspect is more important.

A3. The Ethics of Open Source

Open source is not just about software, but also about how knowledge is shared.

Should All Software Be Open Source?

Advantages:

- Transparency
- Better collaboration

- Faster development

Disadvantages:

- Lack of structured support
- Companies may lose incentives
- Some systems require restricted access

Is It Ethical for Companies to Profit from Open Source?

Companies use open-source tools to build products. This can be ethical if they contribute back to the community. If they only use resources without contributing, it becomes unfair.

Standing on the Shoulders of Giants

Modern software is built on previous work. Git itself is based on earlier ideas, and many tools today depend on Git. Open source allows faster progress by building on existing work.

Conclusion

Git represents the core idea of open source: collaboration, transparency, and shared control.

Part B — Linux Footprint

Installation Method

Git can be installed using package managers such as apt or yum.

```shell
git --version
```

This command verifies that Git is installed.

Key Directories

- `/usr/bin/git` → executable
- `/etc/gitconfig` → system settings
- `~/.gitconfig` → user settings

User and Permissions

Git runs under the current user.

```shell
whoami
```

This improves security and avoids system-level risks.

Service Management

Git is not a service. It runs through commands:

```shell
git init

git clone

git status
```

Update Model

Git is updated through package managers:

```shell
sudo apt upgrade git
```

Conclusion

Git integrates well with Linux and follows system conventions.

Part C — The FOSS Ecosystem

Git is part of a larger open-source ecosystem.

script1_system_identity.sh ● script2_package_inspector.sh ●

script1_system_identity.sh

```bash
#!/bin/bash
# ========================================
# Script 1: System Identity Report
# Author: RESHABH LAHOTI | 24BCE10343
# Description: Displays system information
# ========================================
# --- Variables ---
STUDENT_NAME="= Reshabh Lahoti"
SOFTWARE_CHOICE="Git"
LICENSE="GPL v2"
# --- System Info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime | awk -F'(up |,)' '{print $2}')
CURRENT_DATE=$(date "+%d %B %Y %H:%M:%S")
# Works for both Linux and Mac
DISTRO=$(uname -s)
# --- Display Output ---
echo "========================================"
echo " Open Source Audit - $STUDENT_NAME"
echo "========================================"
echo ""
echo "Chosen Software: $SOFTWARE_CHOICE"
echo ""
echo "System Information"
echo "----------------------------------------"
echo "OS           : $DISTRO"
echo "Kernel       : $KERNEL"
echo "Logged in as : $USER_NAME"
echo "Home Dir     : $HOME_DIR"
echo "Uptime       : $UPTIME"
echo "Date & Time  : $CURRENT_DATE"
echo ""
echo "License Info"
echo "----------------------------------------"
echo "This system runs Unix/Linux based OS."
echo "Git is licensed under $LICENSE."
echo "You are free to use, modify, and share it."
echo ""
echo ""“In open source, you control your own destiny.”""
echo "========================================"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
mac@Devs-MacBook-Air-2 24bce10343 % chmod +x script1_system_identity.sh
mac@Devs-MacBook-Air-2 24bce10343 % ./script1_system_identity.sh
========================================
 Open Source Audit - = Reshabh Lahoti
========================================

Chosen Software: Git

System Information
----------------------------------------
OS           : Darwin
Kernel       : 25.0.0
Logged in as : mac
Home Dir     : /Users/mac
Uptime       :  2:23
Date & Time  : 31 March 2026 22:53:11

License Info
----------------------------------------
This system runs Unix/Linux based OS.
Git is licensed under GPL v2.
You are free to use, modify, and share it.

“In open source, you control your own destiny.”
========================================
mac@Devs-MacBook-Air-2 24bce10343 %
```

script1_system_identity.sh ● script2_package_inspector.sh ●

script2_package_inspector.sh

```bash
#!/bin/bash
# ========================================
# Script 2: FOSS Package Inspector
# Author: Reshabh Lahoti | 24BCE10343
# Description: Checks if a package is installed
# and displays details + philosoph
# ========================================
# --- Package Name ---
PACKAGE="git"
echo "========================================"
echo " FOSS Package Inspector"
echo "========================================"
echo ""
# --- Check if package is installed ---
if command -v $PACKAGE &> /dev/null
then
    echo "[✓] $PACKAGE is installed on this system."
    echo ""
    echo "Version Information:"
    echo "----------------------------------------"
    $PACKAGE --version
else
    echo "[✗] $PACKAGE is NOT installed."
    echo "Install using:"
    echo "brew install $PACKAGE   (Mac)"
    echo "sudo apt install $PACKAGE   (Linux)"
fi
echo ""
echo "Philosophy Note:"
echo "----------------------------------------"
# --- Case Statement ---
case $PACKAGE in
    git)
        echo "Git: Created to remove dependency on proprietary systems."
        echo "It represents control, transparency, and distributed development."
        ;;
    apache2|httpd)
        echo "Apache: A web server that helped build the open internet."
        ;;
    firefox)
        echo "Firefox: A browser focused on privacy and open web standards."
        ;;
    python3|python)
        echo "Python: A community-driven language promoting simplicity and collaboration."
        ;;
    vlc)
        echo "VLC: A media player built openly to support all formats without restriction."
        ;;
    *)
        echo "$PACKAGE: An open-source tool contributing to modern computing."
        ;;
esac
echo ""
echo "========================================"
```

script1_system_identity.sh | script2_package_inspector.sh | script3_disk_audit.sh

script3_disk_audit.sh

```bash
#!/bin/bash

# ========================================
# Script 3: Disk and Permission Auditor
# Author: RESHABH LAHOTI | 24BCE10343
# Description: Checks directory size and permissions
# ========================================

# --- Directories to check ---
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "========================================"
echo " Disk and Permission Audit"
echo "========================================"
echo ""

printf "%-15s %-25s %-10s\n" "Directory" "Permissions (Owner Group)" "Size"
echo "----------------------------------------------------------------"

# --- Loop through directories ---
for DIR in "${DIRS[@]}"
do
    if [ -d "$DIR" ]; then

        # Get permissions, owner, group
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')

        # Get size
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)

        # Print formatted output
        printf "%-15s %-25s %-10s\n" "$DIR" "$PERMS" "$SIZE"

    else
        printf "%-15s %-25s %-10s\n" "$DIR" "Not Found" "-"
    fi
done

echo ""
echo "----------------------------------------"
echo " Git-Specific Check"
echo "----------------------------------------"

# --- Check Git config ---
GIT_CONFIG="$HOME/.gitconfig"

if [ -f "$GIT_CONFIG" ]; then
    PERMS=$(ls -l "$GIT_CONFIG" | awk '{print $1, $3, $4}')
    SIZE=$(du -h "$GIT_CONFIG" 2>/dev/null | cut -f1)

    echo "[✓] Git config found: $GIT_CONFIG"
    echo "Permissions: $PERMS"
    echo "Size: $SIZE"
else
    echo "[✗] No Git config found."
    echo "Run: git config --global user.name \"Your Name\""
fi

echo ""
echo "========================================"
```

```
mac@Devs-MacBook-Air-2 24bce10343 % chmod +x script3_disk_audit.sh
mac@Devs-MacBook-Air-2 24bce10343 % ./script3_disk_audit.sh
========================================
 Disk and Permission Audit
========================================

Directory       Permissions (Owner Group) Size
----------------------------------------------------------------
/etc            lrwxr-xr-x@ root wheel      0B
/var/log        drwxr-xr-x root wheel      50M
/home           lrwxr-xr-x root wheel       0B
/usr/bin        drwxr-xr-x root wheel      80M
/tmp            lrwxr-xr-x@ root wheel      0B

----------------------------------------
 Git-Specific Check
----------------------------------------
[✘] No Git config found.
Run: git config --global user.name "Your Name"

========================================
mac@Devs-MacBook-Air-2 24bce10343 %
```

script1_system_identity.sh | script2_package_inspector.sh | script3_disk_audit.sh | script4_log_analyzer.sh

script4_log_analyzer.sh

```bash
#!/bin/bash

# ========================================
# Script 4: Log File Analyzer
# Author: RESHABH LAHOTI | 24BCE10343
# Description: Counts keyword occurrences in log file
# ========================================

# --- Input Arguments ---
LOGFILE=$1
KEYWORD=${2:-"error"}   # default keyword = error

COUNT=0

echo "========================================"
echo " Log File Analyzer"
echo "========================================"
echo ""

# --- Check if file exists ---
if [ ! -f "$LOGFILE" ]; then
    echo "[✗] File not found: $LOGFILE"
    echo "Usage: ./script4_log_analyzer.sh <logfile> [keyword]"
    exit 1
fi

# --- Check if file is empty ---
if [ ! -s "$LOGFILE" ]; then
    echo "[!] File is empty: $LOGFILE"
    exit 0
fi

echo "Analyzing file: $LOGFILE"
echo "Keyword       : $KEYWORD"
echo ""

# --- Read file line by line ---
while IFS= read -r LINE
do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

# --- Print result ---
echo "----------------------------------------"
echo "Keyword '$KEYWORD' found $COUNT time(s)."

# --- Show last 5 matches ---
echo ""
echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5

echo ""
echo "========================================"
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
mac@Devs-MacBook-Air-2 24bce10343 % chmod +x script4_log_analyzer.sh
mac@Devs-MacBook-Air-2 24bce10343 % nano sample.log
mac@Devs-MacBook-Air-2 24bce10343 % ./script4_log_analyzer.sh sample.log error
========================================
 Log File Analyzer
========================================

Analyzing file: sample.log
Keyword       : error

----------------------------------------
Keyword 'error' found 3 time(s).

Last 5 matching lines:
ERROR: Failed to load module
error: Disk issue detected
ERROR: Timeout occurred

========================================
mac@Devs-MacBook-Air-2 24bce10343 %
```

script1_system_identity.sh | script2_package_inspector.sh | script3_disk_audit.sh | script4_log_analyzer.sh | script5_manifesto.sh

script5_manifesto.sh

```bash
#!/bin/bash

# ==========================================
# Script 5: Open Source Manifesto Generator
# Author: RESHABH LAHOTI | 24BCE10343
# Description: Generates a personalized manifesto
# ==========================================

echo "========================================"
echo " Open Source Manifesto Generator"
echo "========================================"
echo ""

# --- User Input ---
read -p "1. Name one open-source tool you use daily: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. What would you build and share freely? " BUILD

# --- Date and File ---
DATE=$(date "+%d %B %Y")
OUTPUT="manifesto_$(whoami).txt"

echo ""
echo "Generating your manifesto..."
echo ""

# --- Write to File ---
echo "========================================" > "$OUTPUT"
echo " MY OPEN SOURCE MANIFESTO" >> "$OUTPUT"
echo " Generated on: $DATE" >> "$OUTPUT"
echo " By: $(whoami)" >> "$OUTPUT"
echo "========================================" >> "$OUTPUT"
echo "" >> "$OUTPUT"

echo "I believe in the power of open-source software." >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "Every day, I rely on $TOOL — not built for profit alone," >> "$OUTPUT"
echo "but shaped by a community that values shared progress." >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "To me, freedom means $FREEDOM." >> "$OUTPUT"
echo "It is the ability to learn, modify, and build without restriction." >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "One day, I will create $BUILD and share it openly," >> "$OUTPUT"
echo "so others can build on it, just as I have built on the work of others." >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "Open source is not just code — it is collaboration." >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "========================================" >> "$OUTPUT"

# --- Display Output ---
echo "[✓] Manifesto saved to: $OUTPUT"
echo ""
echo "Your Manifesto:"
echo "----------------------------------------"
cat "$OUTPUT"

echo ""
echo "========================================"
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

mac@Devs-MacBook-Air-2 24bce10343 % chmod +x script5_manifesto.sh
mac@Devs-MacBook-Air-2 24bce10343 % ./script5_manifesto.sh
========================================
 Open Source Manifesto Generator
========================================

1. Name one open-source tool you use daily: Git
freedom
a startup platform2. In one word, what does 'freedom' mean to you? 3. What would you build and share freely?

Generating your manifesto...

[✔] Manifesto saved to: manifesto_mac.txt

Your Manifesto:
----------------------------------------
========================================
 MY OPEN SOURCE MANIFESTO
 Generated on: 31 March 2026
 By: mac
========================================

I believe in the power of open-source software.

Every day, I rely on Git — not built for profit alone,
but shaped by a community that values shared progress.

To me, freedom means freedom.
It is the ability to learn, modify, and build without restriction.

One day, I will create a startup platform and share it openly,
so others can build on it, just as I have built on the work of others.

Open source is not just code — it is collaboration.

========================================

========================================
mac@Devs-MacBook-Air-2 24bce10343 %
```

Dependencies

Git is written in C and depends on basic system libraries.

What Git Has Enabled

Git has enabled platforms such as:

- GitHub
- GitLab
- Bitbucket

These platforms support collaboration and project management.

Role in Development

Git is used for:

- Version control
- Code management
- Deployment

Community

Git is maintained by a global community of developers.

Conclusion

Git acts as a foundation for modern software development.

Part D — Open Source vs Proprietary

Comparison Table

Factor	Open Source (Git)	Proprietary
Cost	Free	Paid
Security	Open	Closed
Support	Community	Company
Flexibility	High	Limited

Control	Shared	Centralized

Analysis

Git provides flexibility and transparency. Users can modify it as needed.

Proprietary tools provide structured support but limit control.

Final Verdict

Git is a better choice for most use cases because it is open and flexible. Proprietary tools may still be useful in specific situations.

Personal Position

I would recommend Git for real-world use. It is reliable, widely used, and supported by a strong community.

Conclusion

This project helped me understand that open source is more than just free software. It is about collaboration, control, and continuous improvement.

Git is a strong example of how open-source tools can solve real problems and shape modern software development.